

SIMATS ENGINEERING — ASSESSMENT TOOL 1

Problem Statement — Case Study (Annexure A)

Course Name: Artificial Intelligence **Course Code:** CSA17

Course Outcome (CO3): Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution and chaining methods to perform logical reasoning for drawing valid conclusions and decision making. (BL4)

Assessment Weightage: 50% **Duration:** 1 Hour **Questions:** 4 **Total Marks:** 40

This document consolidates the four case-study problem statements exactly as issued in the assessment brief. Each case study presents a small logical-agent knowledge base and asks the student to apply a specific knowledge-representation or reasoning technique — propositional logic encoding, forward chaining, backward chaining, unification, resolution/CNF conversion, or belief-state construction — to reach a verifiable conclusion.

Case Study 1 — Smart Medical Diagnosis System

A hospital is designing a rule-based Logical Agent for preliminary patient diagnosis. The system's knowledge base contains the following rules and facts:

- i. If a patient has Fever AND Cough $\rightarrow$ Possible Flu
- ii. If a patient has Fever AND Rash $\rightarrow$ Possible Measles
- iii. If a patient has Cough AND Breathlessness $\rightarrow$ Possible Pneumonia
- iv. Facts about Patient A: Fever = True, Cough = True, Breathlessness = True

(a) Represent the above rules and facts precisely using Propositional Logic symbols. Define each symbol used. (*Marks distribution — see rubric*)

(b) Apply Forward Chaining on the knowledge base to derive all possible diagnoses for Patient A. Show each inference step with the rule fired and fact derived.

(c) The goal is to confirm 'Patient A has Pneumonia.' Apply Backward Chaining and draw the goal reduction tree to verify this goal.

Case Study 2 — Autonomous Traffic Management Agent

A smart city traffic controller uses a FOL-based knowledge base to manage emergency vehicle routing:

- I. $\forall x: \text{Vehicle}(x) \wedge \text{EmergencyType}(x) \rightarrow \text{ClearPath}(x)$ [Rule 1]
- II. $\forall x: \text{ClearPath}(x) \rightarrow \text{GreenSignal}(x) \wedge \neg \text{RedSignal}(x)$ [Rule 2]
- III. $\forall x \forall y: \text{Vehicle}(x) \wedge \text{Behind}(x,y) \wedge \text{GreenSignal}(y) \rightarrow \text{Proceed}(x)$ [Rule 3]
- IV. Facts: $\text{Vehicle}(\text{Ambulance}), \text{EmergencyType}(\text{Ambulance}), \text{Vehicle}(\text{CarA}), \text{Behind}(\text{CarA}, \text{Ambulance})$

(a) Apply Unification to match Rule 1 with the given facts. Show the complete substitution set θ (MGU) at each step.

(b) Using the Resolution method, prove that 'Proceed(CarA)' is true. Convert relevant clauses to CNF and show each resolution step.

(c) Analyze whether the current FOL knowledge base is sufficient to handle two simultaneous emergency vehicles. Identify what additional rules or facts are needed and justify logically.

Case Study 3 — Agricultural Expert Reasoning System

An agricultural advisory logical agent operates with the following propositional knowledge base (KB):

- 1) P1: $\text{SoilDry} \rightarrow \text{IrrigationNeeded}$
- 2) P2: $\text{IrrigationNeeded} \wedge \text{CropWheat} \rightarrow \text{ApplyDripMethod}$
- 3) P3: $\neg \text{ApplyDripMethod} \wedge \text{CropWheat} \rightarrow \text{CropAtRisk}$
- 4) Facts: $\text{SoilDry} = \text{True}, \text{CropWheat} = \text{True}$

(a) Convert all sentences P1, P2, P3 and the facts into Conjunctive Normal Form (CNF). Show each conversion step (eliminate implication $\rightarrow$ move $\neg$ inward $\rightarrow$ distribute).

(b) Using Resolution Refutation, prove the theorem 'ApplyDripMethod'. Negate the goal, combine with the KB in CNF, and derive the empty clause $\blacksquare$. Draw the resolution proof tree.

(c) If the fact 'SoilDry' is removed from the KB, analyze step-by-step how logical conclusions change. Does 'CropAtRisk' still follow? Apply appropriate reasoning patterns to justify.

Case Study 4 — Wumpus World Knowledge Agent

An AI agent navigating the Wumpus World perceives: Stench at [1,2], Breeze at [1,1], Glitter at [2,2]. The agent uses a propositional KB with these rules:

- 1) $\text{Stench}[1,2] \rightarrow \text{Wumpus is adjacent to cell } [1,2]$
- 2) $\text{Breeze}[1,1] \rightarrow \text{Pit is adjacent to cell } [1,1]$
- 3) $\text{Glitter}[x,y] \rightarrow \text{Gold is at } [x,y]$
- 4) $\neg \text{Wumpus}[x,y] \wedge \neg \text{Pit}[x,y] \rightarrow \text{Safe}[x,y]$

(a) Construct the agent's belief state from its percepts. Apply Modus Ponens systematically to derive new knowledge facts. List all derived conclusions.

(b) Using Forward Chaining from the agent's percepts, identify the possible locations of the Wumpus and Pit. Show each reasoning step precisely.

(c) Evaluate whether the path $[1,1] \rightarrow [1,2] \rightarrow [2,2]$ is safe for the agent to collect the gold. Use the knowledge derived in (a) and (b) to justify your conclusion with logical inference.

Evaluation Rubric

Criteria	Wt.	L4 – Exemplary	L3 – Proficient	L2 – Developing	L1 – Beginning
Understanding of Case	25	Thorough; all key issues identified	Good; most issues identified	Basic; some issues missed	Poor; problem not identified
Application of Concepts	25	Effectively applies concepts in depth	Applies concepts with minor gaps	Limited application	Fails to apply concepts
Analysis	20	In-depth, strong reasoning	Good, logical reasoning	Basic, limited depth	Weak or no reasoning
Solution Design	20	Innovative, feasible, justified	Logical, adequately justified	Basic, weak justification	Incorrect / no solution
Clarity	10	Clear, well-structured	Organized, understandable	Limited clarity	Poor clarity